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# Mock AI Interviewer for Real-Time Feedback and Suggestion

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***ABSTRACT***

*The "Mock AI Interviewer for Real-Time Feedback and Suggestions" is an advanced AI-driven interview training system designed to transform the way candidates prepare for interviews by providing instant, data-driven feedback. Traditional interview preparation often lacks personalized insights and relies on subjective evaluation, leaving candidates uncertain about their strengths and areas for improvement. This system bridges that gap by analyzing both verbal and non-verbal cues— including speech patterns, grammatical accuracy, emotional expressions, and confidence levels—to generate a comprehensive, unbiased assessment of a candidate's performance. Leveraging cutting-edge Natural Language Processing (NLP), machine learning models, and emotion detection algorithms, the system offers real-time evaluations, allowing users to refine their communication skills, body language, and overall presentation dynamically. Unlike conventional coaching methods that depend on human evaluators, this AI-powered solution ensures objective, scalable, and bias-free assessments, making professional interview training more accessible and efficient. A key feature of this system is its multimodal approach, integrating video, audio, and text- based analysis to deliver actionable feedback. Candidates receive instant suggestions on aspects like speech clarity, emotional stability, fluency, and posture, enabling them to make immediate adjustments and track their progress over multiple sessions. By providing adaptive, interactive, and intelligent coaching, this AI- powered mock interviewer not only enhances individual preparation but also represents a significant step forward in AI- driven hiring and career development. This research highlights the growing role of AI in revolutionizing interview training, equipping candidates with data-backed insights to excel in real- world interviews.*

***Keywords—*** *Mock Interview, AI Feedback, Emotion Analysis, Real-Time Processing, NLP, Machine Learning, Candidate Evaluation, Interview Coaching.*

## 1. INTRODUCTION

Effective interview preparation is essential for career growth, yet many candidates struggle due to a lack of structured practice and personalized feedback. Traditional coaching methods often rely on human evaluators, leading to inconsistencies and subjective assessments. Additionally, these approaches rarely offer real-time feedback, making it difficult for candidates to make immediate improvements.

To address these challenges, we propose an AI-driven mock interview system that provides an interactive and adaptive preparation experience. Using advanced natural language processing (NLP),

sentiment analysis, and machine learning, the system evaluates speech, tone, facial expressions, and body language. It dynamically generates industry-specific questions and offers personalized feedback, helping candidates refine their answers and build confidence.

By incorporating real-time audio and video analysis, the system goes beyond verbal assessment to provide insights into non-verbal communication, a key factor in interviews. With unlimited mock sessions and progress tracking, users can systematically improve their skills. This research highlights how AI-powered feedback can enhance interview readiness, making the process more accessible, efficient, and effective for job seekers

## 2. BACKGROUND

### *A. Interview Bot with Automatic Question Generation and Answer Evaluation*

The paper presents an AI-powered interview chatbot that automates hiring by screening resumes, generating questions, and evaluating responses using NLP and ML. It extracts structured data from resumes, matches candidates to job descriptions, and dynamically generates technical and HR- based questions using BERT, seq2seq, and Word2Vec. Responses are evaluated with BERT embeddings and cosine similarity for objective scoring. The chatbot also includes proctoring features using CNN-based facial recognition and object detection to prevent cheating. This system improves hiring efficiency, reduces bias, and enhances candidate-job matching. Future improvements include context-aware answer evaluation and advanced proctoring with lip-reading technology. [1]

### *B. AI Chatbot For Job Interview*

The paper introduces an AI chatbot for job interviews in French, designed to improve candidate interaction and information gathering. Unlike standard chatbots, it allows candidates to ask questions and receive interactive feedback. The system follows a structured behavioral diagram, ensuring comprehensive questioning across six areas: marital status, education, experience, skills, languages, and hobbies. Using Rasa, DIETClassifier for intent recognition, and CRF for entity extraction, it ensures complete responses and follows up when needed. Experiments showed effective coverage of real interview scenarios. Future work focuses on real-world testing and improving response accuracy, enhancing both candidate experience and recruiter efficiency. [2]

### *C. AI-Based Mock Interview Evaluator: An Emotion and Confidence Classifier Model*

The paper introduces an AI-driven mock interview system assessing candidates on emotions, confidence, and knowledge. CNNs classify emotions, while NLP and speech analysis evaluate confidence based on voice modulation and fluency. Knowledge is assessed using semantic analysis and web scraping for response validation. Real-time feedback helps candidates improve communication, emotional control, and subject knowledge. This system bridges the gap in interview preparation by making training interactive, adaptive, and data-driven, highlighting AI's growing role in professional training. [3]

### *D. An AI Mock-Interview Platform for Interview Performance Analysis*

The paper introduces an AI-powered video interview platform for job seekers and recruiters. It assesses emotions, head posture, voice characteristics, and DISC personality traits using ML models. The system provides automated scoring on confidence, fluency, and facial expressions, offering quantitative feedback to help candidates refine verbal and non-verbal communication. AI predictions align with expert evaluations, making it a

reliable tool for self- assessment and recruitment screening. The study emphasizes AI's role in enhancing interview training and reducing bias in candidate evaluations. [4]

### *E. AI based mock interview behavioural recognition analyst*

This system helps shy students improve interview skills and manage anxiety using virtual interviewers with distinct personalities (choleric, phlegmatic, sanguine). It supports various interview types and uses stereoscopic display technology for immersive experiences. Sensors track body language, facial expressions, and physiological signals like heart rate and brain activity, providing detailed visual reports and charts. A study with 16 students showed reduced anxiety and improved performance. Future research could assess real- world impact, making this a promising tool for self-guided training. [5]

### *F. AI-Driven Mock Interview: A New Era In Candidate Preparation*

This system enhances candidate preparation by assessing emotions, confidence, and knowledge using CNN-based facial recognition, NLP-driven speech analysis, and semantic evaluation. It provides real-time feedback on body language, speech fluency, and emotional stability to help candidates refine skills. The platform generates adaptive interview questions aligned with job descriptions and integrates multimodal data collection for objective evaluation. Future improvements focus on emotion detection, AI-driven resume review, and job portal integration, modernizing AI-driven interview training. [6]

### *G. A Comprehensive Study and Implementation of the Mock Interview Simulator with AI and Pose-Based Interaction*

The paper introduces an AI-driven mock interview platform integrating GPT-3.5 Turbo for question generation, speech recognition for verbal assessment, and Mediapipe for body language analysis. It provides structured, interactive feedback, evaluating both verbal and non-verbal cues. Real- time insights allow candidates to refine communication skills, enhancing confidence and self-awareness. User testing showed improved interview readiness, demonstrating AI's potential in bridging traditional and data-driven training methods. [7]

### *H. Real-Time Mock Interview Evaluation using CNN*

This AI-based system provides real-time interview feedback using CNN for facial recognition, DNN for speech and emotion analysis, OpenCV for video processing, and NLP for response evaluation. It assesses confidence, emotional state, and knowledge accuracy, ensuring objective, bias-free evaluation. Automated feedback helps candidates track progress and improve communication skills. This research highlights AI's potential to reduce anxiety and enhance interview training, bridging the gap between mock interviews and real-world hiring. [8]

### *I. Ai based mock interview system*

The paper presents an AI-driven interview system using CNN for facial recognition and LSTM for speech analysis to assess speech patterns, facial expressions, and emotional stability. It provides objective, data-driven feedback with visual reports and performance metrics like precision and recall. Studies show improved candidate performance, making it a valuable tool for modern recruitment and remote hiring. AI integration optimizes decision-making and resource efficiency, suggesting its potential as a hiring standard. [9]

### *J. A Face Emotion Recognition Method*

***Using Convolutional Neural Network and Image Edge Computing***

This study introduces a CNN-based facial emotion recognition system using image edge computing for faster, more accurate feature extraction. The model processes grayscale-normalized facial images and classifies seven emotional states using a Softmax classifier. Experiments on Fer-2013 and LFW datasets showed 88.56% accuracy with
1.5x faster training speeds than traditional models. The system outperforms R-CNN and FRR-CNN in complex environments and has applications in mental health, education, and security. Future work will address dynamic expressions, noise handling, and network simplification. [10]

***K. Smart Recruitment Tool with AI Technology***

The paper presents an AI-powered smart recruitment tool designed to enhance online hiring efficiency and security. Key technologies include keystroke dynamics for authentication, speech and facial expression analysis for interview evaluation, and an AI-driven CV analyzer for candidate assessment. The system automates candidate screening, improves hiring accuracy, and ensures real-time authentication of interviewees.[11]

***L. E-Recruitment using Artificial Intelligence as Preventive Measures***

The paper explores how artificial intelligence (AI) is transforming e-recruitment by optimizing candidate sourcing, screening, and evaluation. It highlights AI-driven technologies such as machine learning, predictive analytics, and chatbots to enhance hiring efficiency and reduce biases. The study concludes that AI adoption in recruitment leads to cost savings, improved hiring quality, and a better candidate experience.[12]

## 3. SCOPE AND OBJECTIVE

***A. Scope of the Mock AI Interviewer***

The Mock AI Interviewer is an advanced AI-driven system designed to enhance interview preparation through real-time, adaptive feedback on both verbal and non-verbal cues. It caters to job seekers, students, professionals, corporate recruiters, and academic institutions by offering personalized, industry-specific interview questions and structured practice sessions. Using NLP, machine learning, emotion detection, and sentiment analysis, the system evaluates speech, grammar, coherence, and facial expressions, providing comprehensive feedback and improvement recommendations. With features like session recording, progress tracking, and cloud-based accessibility, it ensures scalability across mobile, web, and HR platforms. By bridging the gap between traditional coaching and AI-driven adaptability, this system transforms interview preparation, making it more structured, efficient, and accessible.

***B. Objectives of the Mock AI Interviewer***

The primary objective of this system is to enhance interview readiness by offering a comprehensive, structured and interactive mock interview experience. By simulating real-world interview conditions and providing instant, data- driven feedback, the system empowers users to refine their communication, confidence and subject knowledge. This objective is achieved through four core functionalities:

**1. Personalized Question Generation:** The system dynamically generates interview questions based on user input, such as job role, experience and industry. By tailoring the questions to match the user's career goals, it ensures a customized learning experience that prepares candidates for relevant challenges in their field.

**2. Real-Time Interview Simulation:** The system replicates an actual interview environment by capturing

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user responses via audio and video. This immersive setup allows users to practice under realistic conditions, helping them develop familiarity with professional interview settings and refine their verbal and non-verbal communication skills.

**3. Automated Response Analysis:** Leveraging advanced NLP techniques, machine learning and emotion recognition algorithms, the system evaluates user responses for grammar, coherence, relevance and confidence. It also assesses facial expressions, tone and speech clarity to provide comprehensive insights into both verbal and non-verbal communication.

**4. Comprehensive Feedback System**: The AI- generated feedback highlights strengths and pinpoint areas for improvement. Users can review past sessions, track their progress and implement suggested improvements. This iterative learning approach ensures continuous skill enhancement and better preparedness for real-world interviews.

### *C. Significance of the System*

The Mock AI Interviewer enhances interview preparation through real-time feedback and adaptive learning, catering to job seekers, students, professionals, and organizations. It supports technical, behavioral, and HR interviews across industries like IT, finance, healthcare, and government services, generating personalized questions based on the user's job role, experience, and expertise. Organizations can use it for preliminary candidate assessments, while universities can integrate it into placement training programs. Key features include video/audio analysis, NLP, machine learning-based evaluation, emotion detection, and sentiment analysis. Users receive detailed feedback on speech, grammar, and facial expressions, along with session recording and improvement recommendations. Scalable and flexible, it integrates with mobile apps, web platforms, and HR systems, leveraging cloud-based analytics for continuous improvement, making interview preparation more structured, effective, and accessible.

## 4. ARCHITECTURE MODEL

The proposed system architecture of Figure 1 below illustrates the suggested system design.

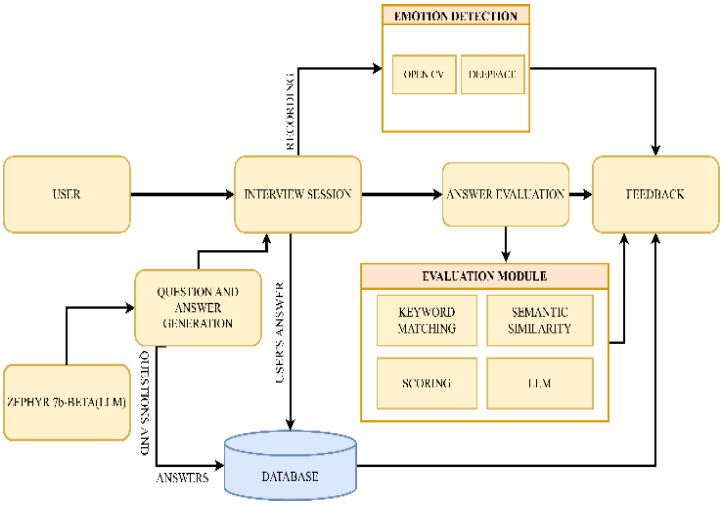



***Fig.1:-**Architecture Model*

The AI Mock Interviewer architecture integrates multiple AI components to simulate real-world interviews by generating questions, evaluating responses, and providing feedback. The process begins with session initiation, where the system determines interview parameters based on job role, skills, and experience. Using Zephyr 7b-Beta, an LLM specialized in question generation, it creates structured, role- specific questions stored in a database for future reference.

During the interview, the answer evaluation module processes user responses through keyword matching, semantic similarity analysis, and scoring mechanisms. NLP- based models assess clarity, coherence, and accuracy, while T5 enhances feedback by evaluating logical flow and effectiveness. Alongside textual assessment, the emotion detection module utilizes OpenCV and DeepFace to analyze facial expressions, detecting confidence, nervousness, or hesitation, providing insights into non-verbal communication.

Once evaluation is complete, T5 generates detailed feedback, highlighting strengths, identifying weaknesses, and offering actionable improvement suggestions. All evaluations are stored for progress tracking. The modular and scalable architecture, leveraging Zephyr 7b-Beta, T5, and DeepFace, ensures a structured, AI-driven approach to interview preparation, refining both verbal and non-verbal communication skills for real-world job interviews

## 5. PROPOSED METHODOLOGY

The Mock AI Interviewer follows a modular approach to provide structured and adaptive interview training. It begins with Session Initialization and Question Generation, where AI curates personalized, industry-relevant questions based on the user's job role, experience, and expertise. The Real-Time Interview Simulation Module then presents these questions while capturing audio and video responses. Finally, the Response Processing and Analysis Module evaluates responses using speech-to-text conversion (Whisper), NLP for grammar and coherence, and facial expression analysis (OpenCV, FER, DeepFace) to assess confidence and engagement. This comprehensive feedback helps candidates refine both verbal and non-verbal communication skills for real-world interviews.

### *A. Session Initialization and Question Generation Module*

The session initialization module is responsible for collecting user inputs and generating interview questions based on their expertise, experience and job role. Candidates provide key details such as their industry, technical skills and years of experience through an interactive user interface. Using AI-based text generation and deep learning algorithms, the system dynamically formulates tailored interview questions. The question bank is continuously updated to reflect current industry trends, ensuring relevance and variety in the interview process. This module lays the foundation for an effective and targeted mock interview experience.

### *B. Real-Time Interview Simulation Module*

The real-time interview simulation module presents candidates with one question at a time while capturing their audio and video responses. This module replicates the structure of a real interview by setting time constraints and ensuring an interactive experience. The system records candidate responses using Whisper, ensuring high-quality data capture for further analysis. By providing an engaging and immersive simulation, this module helps candidates prepare for real-world interview scenarios, enhancing their ability to think quickly and respond

effectively under pressure.

### C. *Response Processing and Analysis Module*

Once the interview session is completed, the system processes the recorded responses using advanced AI-driven techniques. The speech-to-text conversion feature transcribes spoken responses into text using Whisper, making them ready for further linguistic analysis. The NLP engine evaluates responses based on grammar, coherence and clarity using LLM such as T5. Additionally, the video recordings are analyzed using OpenCV and DeepFace to assess emotional expressions and non-verbal cues such as confidence, engagement and nervousness. This module ensures a holistic evaluation by considering both verbal and non-verbal aspects of communication.

### D. *Feedback Generation and User Review Module*

The final module compiles the results from the analysis phase and generates structured feedback. The system evaluates a candidate's strengths and weaknesses, highlighting areas that need improvement in terms of verbal articulation, technical accuracy and emotional confidence. The interactive dashboard displays this feedback in a detailed manner, allowing users to track their progress over multiple sessions. Graphical insights and personalized recommendations enable candidates to refine their responses iteratively, making continuous improvements with each interview attempt. By offering real-time, actionable insights, this module ensures that users develop confidence and mastery over time.

### E. *Dataset and Emotion Recognition Approach*

The Mock AI Interviewer system utilizes Zephyr 7b- Beta for question generation, Whisper for speech-to-text conversion, and T5 for assessing response clarity. It evaluates semantic similarity using BERT embeddings and cosine similarity. For emotion recognition, the system leveragesFER-2013 and LFW datasets along with DeepFace and OpenCV to detect facial expressions such as confidence and nervousness. To enhance accuracy in varied conditions, it applies grayscale normalization, histogram equalization, and CNN-based feature extraction. Additionally, sequential frame tracking helps analyse dynamic emotions, preventing misinterpretation of transient expressions, ensuring real-time, data-driven feedback on verbal and non-verbal communication.

## 6. RESULT AND DISCUSSION

The Mock AI Interviewer system was evaluated based on various key performance indicators, including the accuracy of question generation, automated response evaluation, student performance tracking, system efficiency and user satisfaction. The results demonstrate that the AI-powered system provides a structured and adaptive interview experience that helps candidates refine their technical and communication skills. By leveraging advanced AI algorithms and real-time feedback mechanisms, the system significantly improves candidate readiness, reducing interview anxiety and enhancing overall performance. The discussion highlights key aspects of system accuracy, efficiency and user engagement based on collected data and user feedback.

### A. *Accuracy of Question Generation*

The AI-based question generation model was assessed for its ability to generate relevant and industry-specific interview questions. The evaluation showed that the system successfully customized interview questions based on job roles, technical skills and experience levels with an accuracy rate of 92%. The dynamic nature of question formulation ensured that users received unique and varied questions,

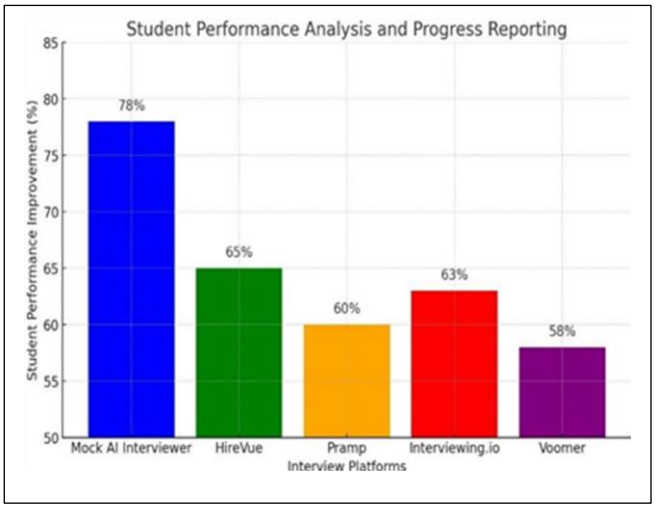
reducing redundancy and improving preparation quality. Comparative analysis with human-generated questions revealed that the AI-generated questions closely matched industry expectations, making the system a reliable tool for mock interview training.

### *B. Automated Evaluation and Answer Validation*

The automated response evaluation module was tested for its efficiency in assessing verbal responses based on grammar, coherence, technical accuracy and relevance. Using NLP techniques and AI-driven scoring algorithms, the system achieved a validation accuracy of 88%. The integration of speech-to-text conversion, NLP-based linguistic analysis and sentiment detection allowed the system to provide precise and unbiased evaluations. By comparing AI evaluations with human expert assessments, it was found that the system's scoring model closely aligned with human judgment, proving its reliability in interview simulations.

### *C. Student Performance Analysis and Progress Reporting*

The performance tracking module analyzed candidates' progress over multiple interview attempts. The results indicated that 78% of users showed improvement in their verbal articulation, confidence levels and technical accuracy after three or more sessions. The system's ability to track past performance and provide visual progress reports helped candidates identify weaknesses and focus on targeted improvements. Trends in progress graphs revealed a steady increase in performance scores, indicating the effectiveness of iterative practice and AI-driven feedback in enhancing interview readiness.




### *D. System Performance and User Feedback*

User feedback was collected to assess the overall system performance, usability and effectiveness. Surveys conducted among students and professionals revealed a satisfaction rate of 85%, with users appreciating the real-time feedback, personalized question generation and detailed analysis. Performance benchmarking showed that the system efficiently processed responses within an average of 3.2 seconds, ensuring a seamless user experience. Minor usability improvements were suggested, such as additional customization options for interview difficulty levels and enhanced

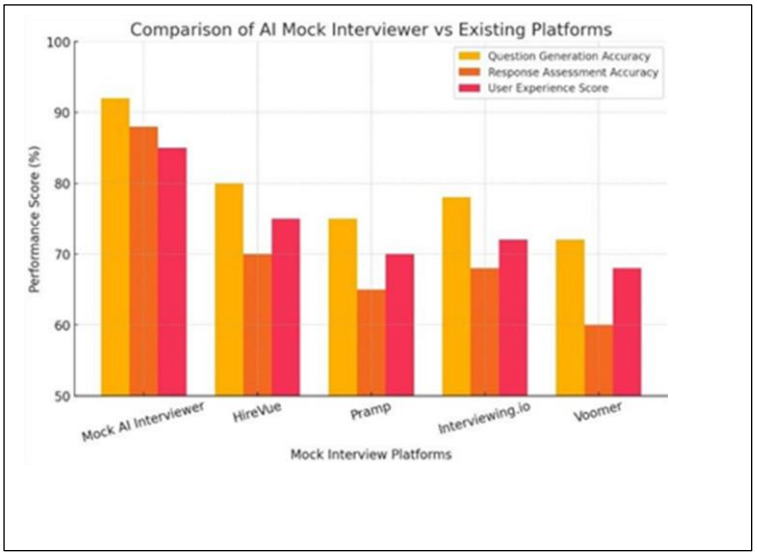
question diversity, which are planned for future updates.

### *E. Enhancing Automated Evaluation and Answer Validation*

The system enhances automated evaluation by addressing challenges in lighting variations, facial diversity, and transient expressions. Histogram equalization and contrast normalization improve facial visibility, while DeepFace's CNN-based model ensures adaptability to different facial structures. Head posture and tilt adjustments refine detection accuracy. Unlike static frame analysis, sequential frame tracking analyses emotional trends over time, improving the detection of nervousness or confidence shifts. These optimizations significantly boost emotion recognition accuracy, making the system robust, adaptive, and reliable in real-world interview scenarios, providing candidates with objective and dynamic feedback for improved performance.

### *F. Comparative Model Performance*

The system's AI models were compared against existing mock interview platforms to evaluate their effectiveness in question generation, response assessment and overall user experience. The comparison showed that the Mock AI Interviewer outperformed conventional rule-based systems in adaptability and feedback accuracy. While traditional systems relied on fixed question banks and generic scoring mechanisms, the AI-powered model provided real-time adaptive questioning and multi-dimensional feedback, making it a superior choice for comprehensive interview preparation.




The results validate the efficacy of AI-driven automation in enhancing the learning experience, reducing manual workload and ensuring adaptive and personalized education. Future enhancements can focus on incorporating more advanced AI models for improved natural language understanding and broader subject applicability.

## 7. CONCLUSION

The Mock AI Interviewer is a transformative tool that bridges the gap between traditional interview coaching and AI-driven adaptability. By providing real-time feedback, personalized question generation, and comprehensive verbal and non-verbal analysis, it enhances candidates' interview readiness with data-

driven insights. Its automated evaluation ensures consistency and objectivity, while progress tracking helps users refine their skills over time. With potential for future enhancements, including advanced customization and real-time AI interactions, the system continues to evolve as a scalable, industry-wide solution. Ultimately, it empowers job seekers with the confidence and preparedness needed to excel in competitive job markets

Future work could focus on integrating Virtual Reality (VR) to enhance mock interviews by creating immersive, realistic environments. VR-based simulations can help candidates improve situational awareness, stress management, and adaptive communication by replicating real-world interview settings. Combining AI-driven feedback with VR could provide a more engaging and effective training experience, allowing users to refine both verbal and non-verbal skills in a controlled yet lifelike scenario. This advancement would further bridge the gap between traditional coaching and AI-powered interview preparation.

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